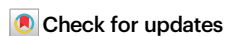


A machine learning framework for predicting and modulating condition-dependent protein phase separation

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Protein phase separation is a fundamental process in organizing membraneless organelles and is implicated in pathological conditions. Importantly, this process is dynamic and depends on conditions such as concentration, temperature, and solvent composition. However, current machine learning models infer phase separation propensity solely from amino acid sequences, failing to capture these context-dependent behaviors. Here we show that LLPSense, a machine learning framework that integrates protein language model embeddings with environmental parameters, achieves accurate, condition-aware predictions of phase separation. Multiple experimental validations confirm LLPSense's predictive power and utility. The model reveals complex, temperature-dependent reentrant behavior in SGTA, previously unrecognized as phase-separating. Moreover, LLPSense accurately predicts mutations in Parkinson's disease-associated α -synuclein that either enhance or suppress phase separation. Beyond predictive accuracy, model-guided mutagenesis enables the modulation of phase behavior. Collectively, LLPSense establishes a robust computational framework for interrogating phase landscapes, facilitating mechanistic disease studies and programmable condensate design.

Biomolecular phase separation of proteins and nucleic acids drives the formation of diverse membraneless organelles and contributes to the spatial organization of biochemical processes within cells¹. Dysregulation of protein phase behavior has been increasingly implicated in the pathogenesis of various diseases, including neurodegenerative disorders^{2–4}. Many proteins have been shown to undergo phase separation, thereby forming membraneless reaction hubs that regulate diverse cellular processes⁵. Despite substantial progress, experimentally identifying phase-separating proteins and characterizing their phase behavior remain labor-intensive and low-throughput. To overcome these limitations, numerous machine learning (ML) approaches have been developed. In particular, several ML predictors^{6–9} have been trained to estimate phase separation

propensity directly from protein sequences, achieving superior performance compared with earlier in silico screening methods¹⁰. Among them, the PSPire¹¹ exhibits remarkable predictive accuracy, even for proteins lacking intrinsically disordered regions (IDRs). Another ML model, PSPHunter¹², identified key residues that drive phase separation and demonstrated that truncation of 6 residues in GATA3 was sufficient to inhibit its phase separation. More recently, catGRANULE 2.0¹³ has shown improved predictive performance relative to these predecessors and was the first to attempt to predict mutants with increased phase-separation propensity.

Previous sequence-based ML prediction models have relied on static binary classification (i.e., whether a protein is capable of phase separation). However, natural liquid-liquid phase separation (LLPS) of

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proteins is a dynamic, stimulus-responsive phenomenon governed by complex physicochemical conditions (e.g., pH, temperature)^{14,15}. Because conventional methods rely exclusively on sequence feature similarity, they inherently treat phase separation as a fixed property, failing to account for the fact that the same protein's behavior varies with conditions. This reliance limits their ability to precisely estimate relative LLPS propensity, which is vital for characterizing the divergent effects of mutations that can either suppress or enhance phase behavior. Therefore, capturing the full spectrum of such modulation requires bidirectional scales that enable precise comparison of phase separation probabilities, supported by a comprehensive dynamic range spanning the global condition space. Droppler¹⁶ represents a pioneering effort to address this limitation by employing a neural network to estimate LLPS probability under various conditions. Despite this advancement, its predictive performance remains limited, thereby restricting broader applicability. The authors of Droppler acknowledged that modeling condition-dependent LLPS is substantially more challenging than sequence-only classification. Consequently, developing a robust predictor that integrates both intrinsic protein sequences and extrinsic conditional variables is essential for the scalable computational analysis of LLPS behavior across large protein datasets¹⁵. Such a condition-aware framework would provide a powerful platform for investigating LLPS biology and for enabling the in silico design of programmable biomolecular condensates.

In this study, we present LLPSeSense, a condition-aware ML framework for predicting protein phase behavior across diverse physicochemical environments. LLPSeSense accurately estimates the phase-separation probability of a given protein sequence under a wide range of parameters, including protein concentration, temperature, pH, crowding agents, salt concentration, and glycerol content. We thoroughly validated the predictive power and experimental utility of LLPSeSense through multiple analyses. Initially, we challenged the model to screen for overlooked phase-separating proteins within datasets previously classified as negative. LLPSeSense flagged 16.4% of these sequences as potential phase-separating candidates, and this predicted subset included proteins already reported in the literature to undergo phase separation. Notably, this screen identified SGTA as an unrecognized LLPS protein, whose complex reentrant, temperature-dependent phase behavior was accurately predicted and subsequently confirmed experimentally. LLPSeSense also pinpointed specific residues in Parkinson's disease-associated α -synuclein, predicting substitutions that either enhance or suppress phase separation. Comparative analysis with a sequence-only predictor confirmed that the condition-aware framework facilitates mutational analysis at single-residue resolution and bidirectional prediction. We additionally demonstrated that the conditional dependence of LLPS can be modulated by altering intrinsic sequence properties. Specifically, mutational engineering of the lower critical solution temperature (LCST)-type protein UBQLN4 shifted its phase behavior toward an upper critical solution temperature (UCST)-like regime, highlighting that LLPSeSense can guide sequence modifications for tuning condition-specific phase behavior.

Results

Protein language model embeddings facilitate accurate prediction of phase separation

Motivated by the recent success of applying language-based encoding methods, such as Word2Vec, to LLPS prediction^{7,12}, we hypothesized that leveraging advanced pretrained protein language models (pLMs) could provide a more accurate and efficient framework. Among available pLMs, we selected ProtT5¹⁷ to extract high-quality sequence embeddings for both training and inference, as prior work (Seq2Phase)¹⁸ has demonstrated that pLM-derived embeddings enable accurate prediction of proteins recruited to biomolecular condensates. Before integrating environmental variables, we validate this pLM-based strategy using the conventional sequence-based, binary

LLPS prediction task. We selected PSPire as a comparative reference because, in addition to its high predictive performance near the completion of our model, it demonstrated improved identification of phase-separating proteins lacking intrinsically disordered regions (IDRs). This distinction is crucial for assessing whether pLMs can provide generalized protein representations that capture phase separation properties across the full structural spectrum. Accordingly, we trained and evaluated our models using the PSPire dataset, which categorizes human proteins into three groups: intrinsically disordered phase-separating proteins (ID-PSPs), structured phase-separating proteins (noID-PSPs), and non-phase-separating proteins (non-PSPs). For binary classification, ID-PSPs and noID-PSPs were treated as positive samples, while non-PSPs were considered negatives.

We developed a gradient-boosted decision tree classifier based on XGBoost¹⁹. A previous study demonstrated that tree-based classifiers, such as XGBoost and Random Forest, outperform neural-network architectures and other ML approaches for LLPS prediction¹². We first obtained pLM-based embeddings using ProtT5, which encodes an amino acid sequence of length N into an $N \times 1024$ embedding matrix. Mean pooling was then applied across the sequence dimension to obtain a single 1024-dimensional vector representation. These averaged embeddings were subsequently used as input features for model training and evaluation, yielding a model we termed LLPSeq (Fig. 1a). This model demonstrated remarkable predictive performance. In both the ID and noID test sets, the area under the receiver operating characteristic curve (AUROC) was comparable to values reported for PSPire, while the area under the precision-recall curve (AUPRC) surpassed it (Fig. 1b–e). To benchmark protein language embeddings against conventional feature-based models with a similar architecture, we constructed a baseline model trained on 55 knowledge-based engineered features (e.g., molecular weight, net charge, hydrophilicity, and low-complexity region content) widely used in previous studies^{6,7}. The overall performance of this baseline model was inferior to that of PSPire and LLPSeq, especially in the IDR-lacking (noID) test set (Fig. 1d, e). These findings indicate that, even without handcrafted feature engineering, embeddings derived directly from pretrained protein language models provide a more effective and generalizable representation for LLPS prediction.

LLPSeSense accurately predicts condition-dependent protein phase behavior

Building on the promising performance of the pLM-based LLPSeq model, we extended this framework to develop a comprehensive predictor that integrates both intrinsic sequence information and extrinsic environmental conditions. A significant bottleneck in developing condition-aware models has been the limited availability of large, standardized datasets suitable for machine learning. Although LLPSeSense v2²⁰ provides a broad collection of in vitro LLPS observations, substantial curation is required to render it amenable to model training. Therefore, we assembled a rigorously curated, ML-ready dataset by filtering LLPSeSense v2 specifically for single-protein entries.

For each entry, we extracted and standardized a comprehensive set of 13 condition variables, including protein concentration, temperature, pH, six representative categories of crowding agents, concentrations of three representative salts (NaCl, KCl, and MgCl₂), and glycerol percentage. These 13 variables were sufficient to capture the majority of experimentally annotated entries. In contrast, complex cosolutes such as heparin or RNA were excluded due to data sparsity and mechanistic complexity. Collectively, these selected parameters capture key physicochemical modulators of LLPS, encompassing solvent polarity, ionic strength, and macromolecular crowding. Furthermore, to accommodate the continuous nature of experimental phase diagrams, we implemented a systematic interval-partitioning strategy. This method subdivides reported ranges into uniformly spaced samples, thereby augmenting the effective number of training instances

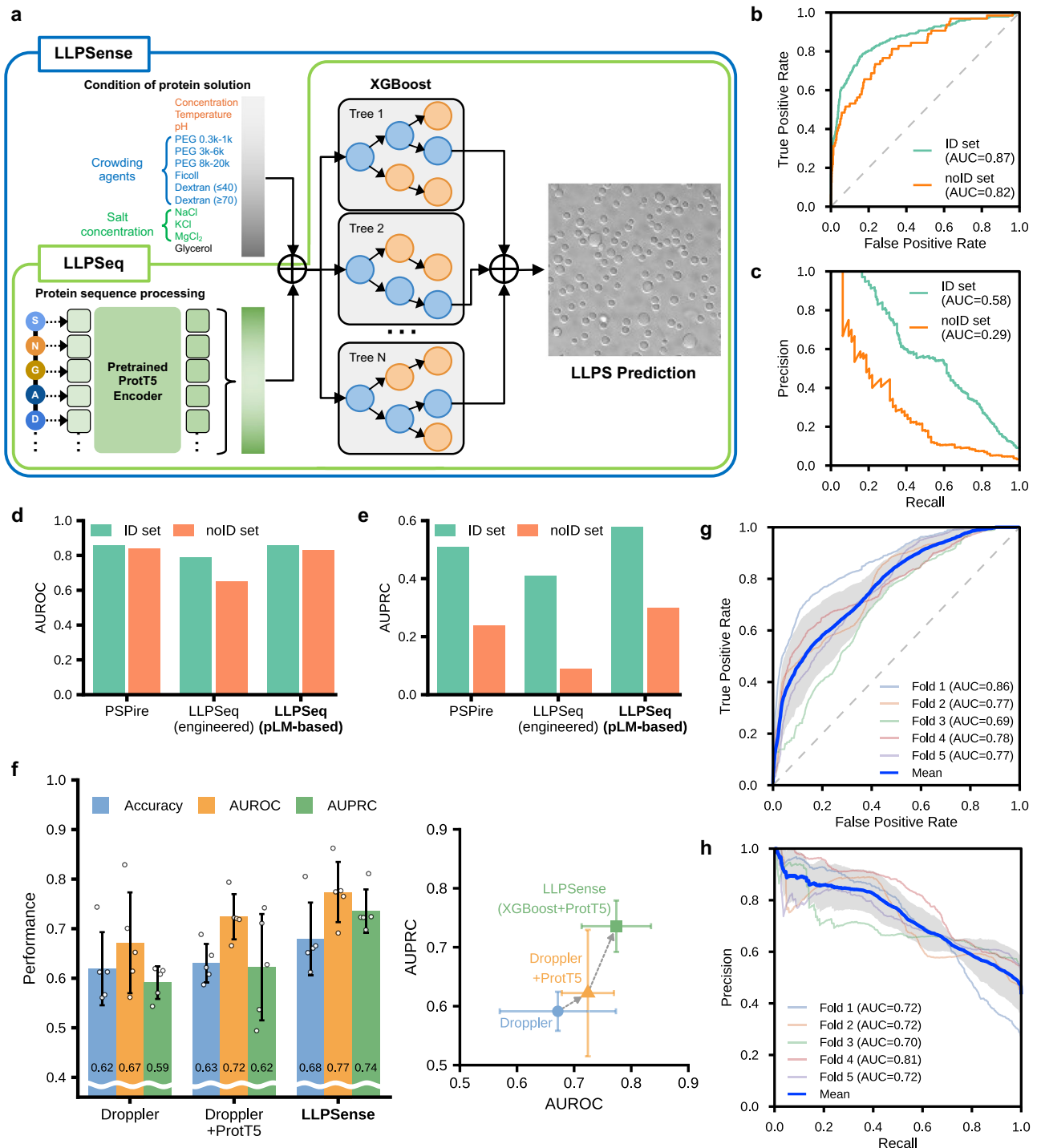


Fig. 1 | Framework and performance evaluation of LLPSeq and LLPSense.

a Schematic overview of the LLPSeq and LLPSense architectures. **b, c** Predictive performance of the protein language model (pLM)-based LLPSeq, evaluated separately on intrinsically disordered phase-separating proteins (ID set) and structured phase-separating proteins (noID set) from the PSPire dataset, is shown as ROC curves (**b**) and PR curves (**c**). **d, e** Bar plots comparing AUROC (**d**) and AUPRC (**e**) of pLM-based LLPSeq with PSPire and an engineered-feature baseline (LLPSeq engineered), evaluated on the same ID and noID sets. **f** Performance comparison of LLPSense

against Droppler and Droppler+ProtT5 (Droppler retrained with ProtT5 embeddings) on the condition-dependent LLPS prediction task, shown as bar plots and an AUROC–AUPRC scatter plot. Bar heights and scatter plot points represent mean values across $n = 5$ cross-validation folds; error bars represent ± 1 s.d.; individual dots in the bar plots represent the value from each fold. **g, h** ROC curves (**g**) and PR curves (**h**) for LLPSense from the same 5-fold cluster-based cross-validation. The bold blue lines represent the mean curves across the 5 folds, and the shaded regions represent ± 1 s.d. Source data are provided as a Source Data file.

while preserving experimental validity (see Methods for details). All curated datasets and preprocessing scripts used in this study are publicly available to facilitate future machine learning research on condition-dependent protein phase separation (see Data Availability).

Next, sequence embeddings derived from ProtT5 were concatenated with the normalized condition variables to construct an integrated feature matrix. This combined representation served as the input for an XGBoost-based architecture, and we designated this

predictive framework as LLPsense (Fig. 1a). This multimodal integration enables the model to simultaneously capture intrinsic sequence determinants of LLPS and extrinsic physicochemical factors that modulate condensate formation. We employed a 5-fold cluster-based cross-validation (CV) strategy to assess the generalizability of LLPsense and identify optimal hyperparameters. To prevent overestimation of performance due to sequence homology between training and test sets, proteins were grouped into 234 distinct clusters using MMseqs2²¹, ensuring that sequences in different clusters shared <20% identity at 90% coverage. All proteins within the same cluster were assigned to the same CV fold, thereby maintaining strict separation between training and validation sets. Under this cluster-based 5-fold CV, LLPsense demonstrated robust predictive performance, achieving an average AUROC of 0.77 and an AUPRC of 0.74 (Fig. 1f–h). This performance significantly surpasses the AUROC value reported for Droppler (0.64). Furthermore, even when Droppler was retrained on our expanded dataset, LLPsense continued to exhibit superior predictive accuracy (Fig. 1f, and Supplementary Fig. 1).

Finally, to investigate the effectiveness of each core component of LLPsense, ProtT5 sequence embedding, and model architecture, we conducted a disentanglement experiment. We replaced the sequence encoder of Droppler, a neural network-based model, with ProtT5-derived representations and evaluated the resulting model (Supplementary Fig. 2). Although this modification yielded improved performance relative to the original Droppler, it still underperformed relative to LLPsense, suggesting that ProtT5 sequence embeddings reach their full predictive potential when coupled with an appropriate model architecture (Fig. 1f). Alternative classifiers, including naive Bayes and support vector machine, similarly failed to match the performance of LLPsense (Supplementary Table 1).

Altogether, these results demonstrate that our integrated modeling strategy provides an effective and reliable framework for predicting condition-dependent phase separation. Given that modeling conditional dependencies is significantly more challenging than sequence-only prediction, the achieved performance marks a pivotal milestone in the field. We next investigated whether this *in silico* predictive accuracy translates into practical utility in real experimental settings.

Identification of potential phase-separating proteins in the previously classified negative dataset

We screened the human proteome dataset previously classified as non-PSP (used in the PSPire study) using LLPsense. Remarkably, a substantial fraction of proteins (16.4%, 1,681/10,252) were predicted to undergo phase separation across a scanned temperature range of 10–50 °C under physiological conditions (100 μ M protein concentration, pH 7.3, and 160 mM NaCl; Fig. 2a). Among these proteins, SS18²², FOXF2²³, and Atrophin-1²⁴ (Fig. 2b) have been reported to exhibit phase-separating behavior, despite their inclusion in the non-PSP set. Notably, SGTA stands out among the identified candidates for its predicted temperature-dependent reentrant phase behavior, which displays characteristics of both the UCST and LCST regimes (Fig. 2c).

To experimentally validate LLPsense's predicted reentrant phase behavior of SGTA, the protein was recombinantly expressed, and its phase behavior was assessed using solution turbidity measurements, a commonly used quantitative indicator of phase-separation propensity. We observed a PEG concentration-dependent increase in SGTA turbidity at 350 nm (A_{350}), confirming its intrinsic ability to phase separate (Fig. 2d). Consistent with the predicted reentrant profile, turbidity increased modestly at 30–35 °C but showed a much greater increase at both low (4 °C) and elevated (≥ 40 °C) temperatures (Fig. 2e). Microscopic images of SGTA condensates corroborated these findings, showing that the total volume of condensates was larger at elevated temperatures (Fig. 2f). Collectively, these results demonstrate that LLPsense not only identifies unrecognized phase-separating

candidates but also captures their intricate, condition-dependent phase behavior.

To further assess the predictive applicability of LLPsense, we randomly selected three proteins (ASCL1, FBLL1, SEMG2) from candidates predicted to exhibit a phase separation threshold within the 10–50 °C range (Supplementary Fig. 3) and evaluated their behavior *in vitro*. ASCL1 and FBLL1 aggregated during dialysis against PBS (pH 7.3, 160 mM NaCl), resulting in final concentrations below 10 μ M. FBLL1 is known to localize within the nucleolus²⁵, suggesting that it may participate in the phase behavior of this membraneless organelle. SEMG2 is a semenogelin protein responsible for the gelation of seminal fluid²⁶. Upon addition of PEG8k (PEG8000), SEMG2 displayed marked turbidity (Supplementary Fig. 4), consistent with macromolecular condensation.

The aggregation of ASCL1 and FBLL1 may reflect their high intrinsic LLPS propensity, as phase-separating proteins often exhibit marginal solubility and a greater tendency to aggregate^{27,28}. Furthermore, we observed a concentration-dependent transition in the material state: SGTA formed large liquid-like droplets at 160 μ M (with 10% PEG8k), whereas small, gel-like condensates formed at lower concentrations (Fig. 2f). SEMG2 also formed interconnected gel-like condensates rather than canonical liquid droplets (Fig. 2g). Despite extensive research, the criteria for distinguishing liquid-like condensates from percolated gels and irreversible aggregates remain poorly defined²⁹. Accordingly, while LLPsense effectively predicts condensation propensity, it does not yet discriminate among distinct material states. With the expanding understanding of biomolecular phase behavior, LLPS, gelation, aggregation, and fibrillization are viewed not as discrete end states but as interconnected regimes within a shared energy landscape^{30–32}. Our preliminary tests with ASCL1, FBLL1, SEMG2, and SGTA illustrate this continuum, highlighting both the broad predictive reach of LLPsense and the necessity for future data refinements that account for the full liquid–gel–solid spectrum.

Taken together, these predictive and experimental findings underscore a critical limitation in previous methods for constructing negative datasets. Such issues arise primarily because a definitive list of proteins strictly incapable of LLPS has not been established. As a result, previous studies have resorted to heuristic negative sets derived from indirect criteria, such as subtracting known LLPS-positive proteins from the proteome. However, the boundary between LLPS-competent and LLPS-incompetent proteins is inherently fluid, varying across environmental conditions and experimental contexts. Seeking to clarify whether LLPS competence can be unambiguously assigned to individual proteins, we tested two widely used solubility-enhancing fusion tags, SUMO3 and MBP, which are routinely employed in recombinant protein expression to improve yield³³. LLPsense predicted that both proteins would not undergo phase separation in the absence of a crowding agent, but would do so in the presence of 10% PEG8k–20k (Supplementary Fig. 5a). Turbidity measurements and microscopic analysis confirmed that phase separation indeed occurred under 10% PEG8k conditions; in particular, SUMO3 condensates displayed a liquid-like droplet morphology (Supplementary Fig. 5b). The fact that even highly soluble proteins can undergo phase separation under specific conditions, and that LLPsense successfully captures this behavior, is noteworthy. This ambiguity in the criteria for defining LLPS-positive proteins complicates model training and further emphasizes the need for condition-aware strategies.

Validation of LLPsense under diverse environmental conditions

Following our investigation into temperature-dependent regulation, we screened the human proteome across varying NaCl concentrations, which identified a predominantly negative correlation between ionic strength and LLPS propensity (Supplementary Figs. 6, 7). This suppressive trend is consistent with the established principle that salt-dependent LLPS regulation is primarily driven by electrostatic

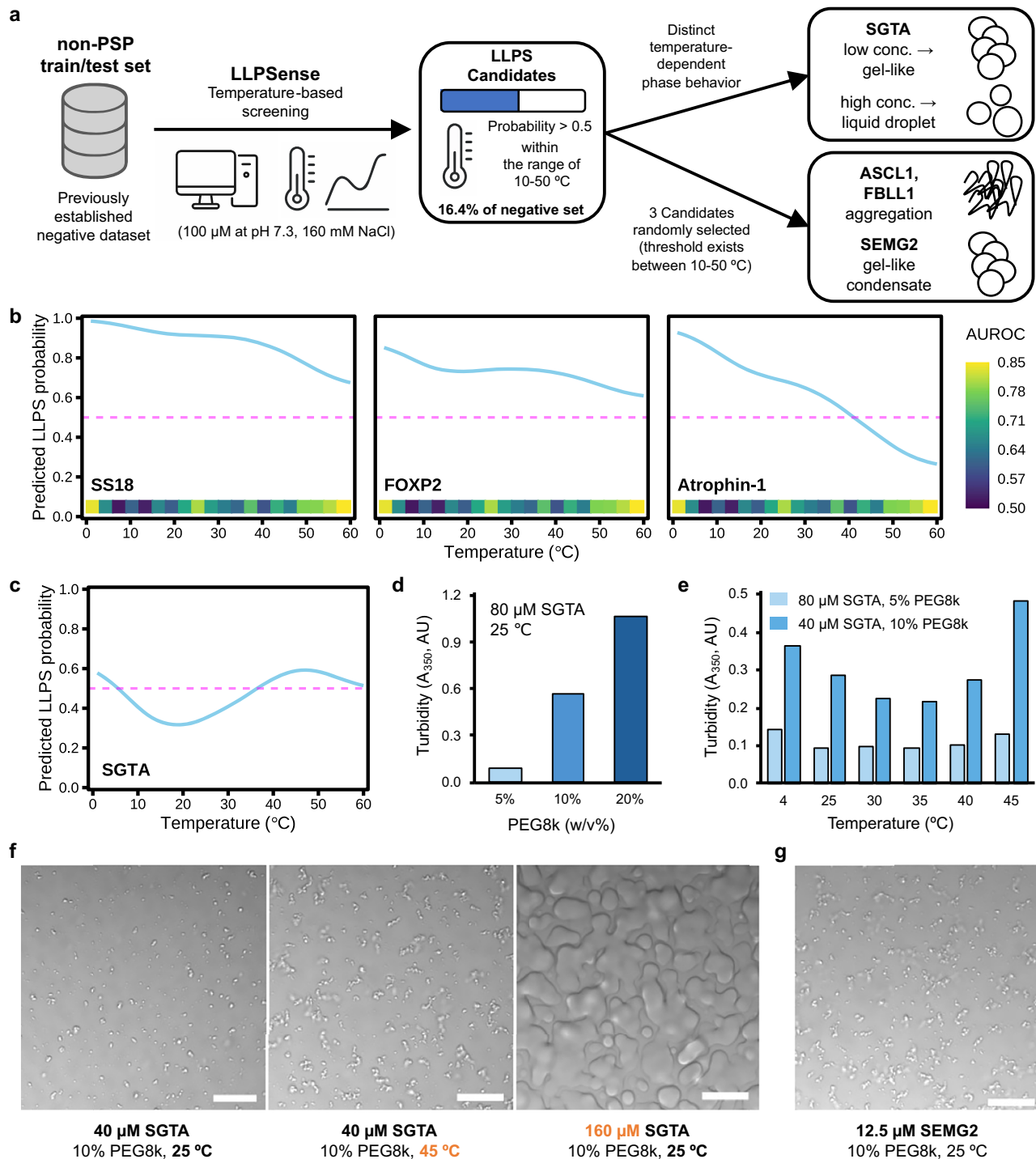


Fig. 2 | LLPSense-guided discovery and experimental validation of LLPS candidates. **a** Workflow for LLPSense-based screening and experimental validation. LLPSense re-evaluated a previously established non-PSP dataset, identifying 16.4% of proteins as potential LLPS candidates (probability > 0.5 within 10–50 $^{\circ}$ C). **b** Predicted temperature profiles of literature-validated phase-separating proteins misclassified in the non-PSP set. Curves were smoothed using a centered moving average with a 15-point window (15 $^{\circ}$ C span). Color bars beneath each curve indicate local prediction reliability, represented as AUROC computed within successive 3 $^{\circ}$ C intervals across the temperature range. **c** Predicted temperature profile of SGTA (smoothed as in **b**),

exhibiting distinctive reentrant phase behavior characterized by mixed UCST/LCST regimes. **d** Turbidity (A_{350}) of SGTA at increasing PEG8000 (PEG8k) concentrations at 25 $^{\circ}$ C. **e** Temperature-dependent turbidity of SGTA under two combinations of protein and PEG8k concentrations, validating the predicted reentrant phase behavior. **f** Representative microscopy images of SGTA condensates demonstrating temperature- and concentration-dependent morphological changes. **g** Microscopy image of SEMG2 condensates forming interconnected gel-like clusters. **f, g** Scale bars, 20 μ m; similar results were confirmed in duplicate independent biological samples, with consistent morphology observed across multiple fields within each sample.

screening³⁴. Elevated ionic strength attenuates the charge–charge interactions essential for condensate formation, while reentrant phase separation could emerge at extreme ionic conditions³⁵. Building on this general trend, we aimed to uncover a protein with an atypical salt-

dependent profile that remains experimentally uncharacterized. LLPSense predicted that RAD23A, a protein with previously reported condensate formation^{36,37} but absent from our training set, would undergo phase separation preferentially at intermediate salt

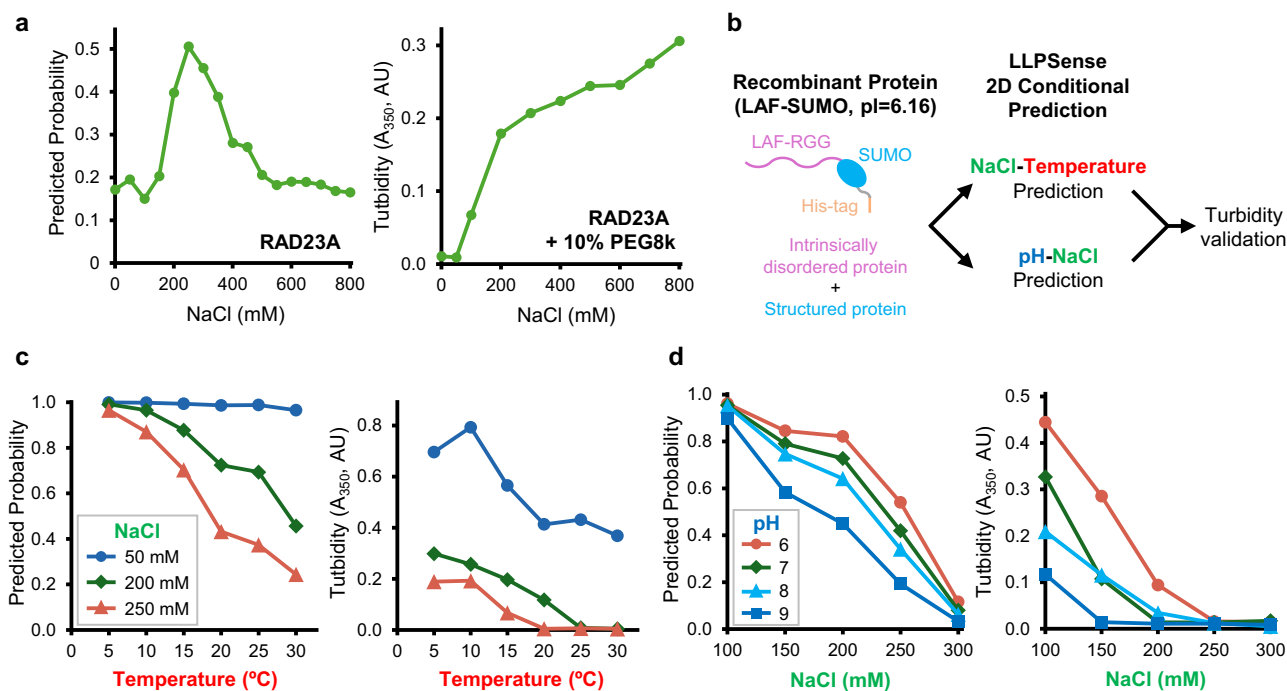


Fig. 3 | Validation of LLPsense predictions under diverse physicochemical conditions. **a** Salt-dependent LLPS prediction (left) and turbidity validation (right) of RAD23A. Turbidity was measured in the presence of 10% PEG8k as a crowding agent. (100 μM RAD23A, pH 8, 25 $^{\circ}\text{C}$) **b** Schematic of the LAF-SUMO recombinant fusion protein and the LLPsense 2D conditional prediction workflow for NaCl-Temperature and pH-NaCl axes. **c** Predicted temperature profiles of LAF-SUMO under varying NaCl concentrations (left) and corresponding turbidity

measurements (right), demonstrating heightened temperature sensitivity under low-salt conditions. (50 μM LAF-SUMO, PBS, pH 7.3, supplemented with 10% glycerol) **d** Predicted NaCl-dependent LLPS probability of LAF-SUMO across pH values (left) and corresponding turbidity measurements (right), showing progressive suppression of phase separation with increasing pH (50 μM LAF-SUMO, 25 $^{\circ}\text{C}$, PBS for pH 6 and 7, TBS for pH 8 and 9; glycerol-free condition). Source data are provided as a Source Data file.

concentrations (Fig. 3a, left). In agreement with the model's predicted trend, turbidity measurements showed that RAD23A remained in a mixed state at low ionic strength (0–50 mM NaCl) but initiated phase separation above 100 mM (Fig. 3a, right). However, contrary to the prediction, turbidity continued to increase at NaCl concentrations exceeding 300 mM. We attribute this deviation to data scarcity for high-NaCl conditions in the currently available database (Supplementary Fig. 8b); consequently, the present model has limited reliability in such high-ionic regimes. Nonetheless, the identification of RAD23A as a protein displaying irregular salt-dependent behavior reaffirms the utility of LLPsense in searching for proteins with specific condition-dependent LLPS trends.

We also designed and expressed a recombinant fusion protein, LAF-SUMO, to evaluate LLPsense's ability to predict LLPS behavior across multiple physicochemical axes. The intrinsically disordered RGG domain of LAF-1 was selected as the phase-separating module for several reasons: LAF-1 is a well-characterized LLPS driver in P granules³⁸, its RGG domain³⁹ can be recombinantly expressed with high stability in vitro, and its high charge density renders it acutely sensitive to both ionic strength and pH, making it an ideal reporter for two-dimensional conditional response mapping. This IDR was fused to an acidic, structured SUMO solubility tag; the resulting hybrid construct differs from the native LAF-1 RGG domain in both charge distribution and structural content, providing a more stringent and compositionally diverse test case (Fig. 3b). The model correctly predicted that LAF-SUMO exhibits heightened temperature sensitivity and a markedly lower dissolution temperature under elevated salt concentrations (Fig. 3c), recapitulating the destabilizing interplay between charge screening and thermal fluctuations. Additionally, LLPsense captured the progressive suppression of phase separation as pH increased above the protein's isoelectric point (pI = 6.16), consistent

with net negative charge accumulation reducing the driving force for condensation (Fig. 3d). These results provide compelling examples of LLPsense's predictive capability across cooperative, multi-axis modulation of protein phase behavior within physiologically relevant ranges of temperature, salinity, and pH.

In silico mapping of the α -synuclein mutational landscape at single-residue resolution

Many diseases are associated with mutations that induce aberrant enhancement^{40–43} or suppression^{44,45} of phase separation. However, relying solely on sequence information poses a significant risk of overfitting to limited positive datasets, which often lack environmental context. Furthermore, such approaches may struggle to capture the relative phase-separating capacity of different sequences, as all proteins validated to undergo LLPS are typically treated as uniform positive instances. Our condition-aware framework addresses these limitations by providing the granularity needed to distinguish sequence behavior under varying environments. To illustrate this capability, we examined α -synuclein, whose LLPS behavior is increasingly recognized as a critical intermediate state preceding pathological fibrillization⁴⁶. By systematically substituting each of the 140 residues of α -synuclein and computing the resulting changes in LLPS probability, we predicted specific mutations that modulate its phase-separation landscape.

LLPsense enabled rapid probability screening across the entire mutational landscape, comprising all possible single-point substitutions (140 residues $\times$ 19 variants; Fig. 4a). To experimentally validate the predicted effects, we selected two mutations projected to exert the strongest influence on phase separation (Fig. 4b). Specifically, the substitution of glutamate with methionine at position 28 (E28M) within the amphipathic N-terminal domain yielded the highest

predicted increase in LLPS probability; we termed this the 'E1 variant'. Conversely, the substitution of valine with aspartate at position 77 (V77D) within the hydrophobic non-amyloid component (NAC) domain resulted in the most significant reduction, designated the 'S1 variant'. By comparison, mutations within the acidic C-terminal tail elicited relatively minor fluctuations in predicted phase behavior.

Using the E1 and S1 variants as scaffolds, we performed iterative rounds of prediction to identify additional mutations that confer additive effects while retaining previously mutated positions. Through this recursive procedure, we constructed a series of variants with multiple mutations, designating the phase-separation enhanced set as E2, E4, and E8, and the suppressed set as S2 (Supplementary Table 2). For these variants, the numerical suffix indicates the cumulative number of incorporated mutations. In addition, to verify that LLPSense accurately predicted E1 based on the specific position (E28) and residue (E-to-M), two control mutants (E28P and E130M) were selected for their lowest predicted LLPS probabilities among possible combinations (Fig. 4c). E28P represents a different substitution at the same position, while E130M represents the identical E-to-M substitution at a different position. Both control mutants exhibit predicted probability significantly lower than that of the E1 variant and slightly below the WT baseline (Fig. 4d).

We expressed and purified all α -synuclein variants under identical conditions and assessed their phase separation behavior by measuring turbidity across nine combinations of PEG8k and protein concentrations. Heatmap visualization of Δ Turbidity, calculated by subtracting WT turbidity from that of each variant's turbidity, revealed a stark contrast between the enhanced and suppressed variant sets (Fig. 4e, and Supplementary Fig. 9). Summing the Δ Turbidity values across all nine conditions yielded the following order: E2 > E1 > WT (baseline) > E28P > E130M > S1 > S2 (Fig. 4f). This ranking was in excellent agreement with LLPSense predictions (Fig. 4d), with the only minor deviation occurring between E130M and E28P, where predicted differences were marginal. Consistent with these results, microscopic analysis of the α -synuclein variants confirmed that those with higher predicted propensities exhibited a clear increase in condensate formation (Supplementary Fig. 10). During PBS dialysis, the hyper-enhanced variants E4 and E8 exhibited pronounced aggregation and structural instability (Supplementary Fig. 11a, b). This observation corroborates our earlier discussion that condensation-prone sequences often exhibit a concomitant increase in aggregation propensity. Nonetheless, both variants still showed turbidity levels higher than WT (Supplementary Fig. 11c). Taken together, these results demonstrate that LLPSense accurately identifies mutations that modulate phase separation and can reliably pinpoint sequence alterations that shift α -synuclein toward LLPS-prone states.

Condition-aware mutational screening affords a bidirectional scale and granular resolution of LLPS propensity

To demonstrate the enhanced accuracy and granularity of our condition-aware framework, we also compared the mutational landscapes of α -synuclein predicted by LLPSense and the sequence-based predictor LLPSeq. Despite the absence of α -synuclein from its training dataset, LLPSeq predicted a high probability of 0.78 for WT, a result likely driven by overfitting to the homologous β -synuclein sequence. Consequently, LLPSeq classified most single-point mutations as suppressive (90.2%; 2039/2260). In contrast, even with the target protein included in its training, LLPSense identified a substantial fraction of mutations as enhancing (24.0%; 543/2260) (Fig. 5a, b), demonstrating a broad bidirectional prediction range. LLPSeq also predicted a monotonic decrease in phase separation potential for enhanced mutants (E1–E8), showing an increasingly pronounced discrepancy from the intended cumulative modulation (Supplementary Fig. 12). In addition, LLPSense can circumvent this saturation issue through its environmental sensitivity. By modulating parameters such as protein concentration or crowding agent levels, the model can shift the baseline

into a neutral regime, near a predicted probability of 0.5, even if a sequence exhibits high intrinsic probability. This capability ensures a sufficient dynamic range to resolve subtle mutational effects that are otherwise masked in static, sequence-only models.

We additionally benchmarked LLPSense on α -synuclein variants previously reported as pathogenic and characterized for LLPS propensity using turbidity and droplet-size measurements^{46–49}. Based on prior experimental observations, these mutations were categorized into four distinct groups: significantly enhanced, mildly enhanced, comparable to WT, and suppressed. LLPSense demonstrated high qualitative consistency with these experimental groupings by accurately reflecting the relative rankings and propensity shifts across all tested substitutions (Fig. 5c). The absolute probability for the WT sequence (0.204) may be subject to moderate overestimation due to its inclusion in the training set, though this is expected to be mitigated as the dataset's diversity and size increase. Nevertheless, the relative comparisons and predicted trends across various mutants remain highly valid, demonstrating that LLPSense accurately reflects the impact of each substitution at single-residue resolution. In contrast, while an existing bidirectional mutation model, catGRANULE 2.0, captured general tendencies, it incorrectly predicted the relative order of certain variants (Supplementary Fig. 13). It also predicted trends for our discovered S1 and E1 variants that were opposite to the experimental observations, which illustrates the difficulty of accurately predicting subtle mutational impacts (Supplementary Table 3).

LLPSense-guided mutations reprogram UBQLN4 from LCST to UCST phase behavior

Most phase-separating proteins exhibit upper critical solution temperature (UCST) behavior, where phase separation is favored at lower temperatures due to reduced entropic costs. In contrast, certain families, such as elastin-like polypeptides (ELPs)⁵⁰ and the UBQLN family⁵¹, exhibit LCST behavior, undergoing phase separation as temperature increases. Our curated dataset included ELPs and UBQLN2, but excluded UBQLN1 and UBQLN4. Notably, while UBQLN1 shares high sequence homology with UBQLN2 (76.21% identity, 82.79% similarity), UBQLN4 is more distantly related (58.72% identity, 70.19% similarity). When the Droppler model was retrained on our dataset, it correctly classified the highly homologous UBQLN1 as an LCST-type protein but failed to identify the more divergent UBQLN4 (Supplementary Fig. 14a). By contrast, LLPSense accurately predicted the LCST behavior of both UBQLN1 and UBQLN4 (Fig. 6a), demonstrating improved generalizability and sensitivity to the sequence determinants underlying temperature-dependent phase behavior.

To experimentally validate these predictions, we examined UBQLN4 *in vitro* and observed a clear increase in turbidity and condensate formation with increasing temperature, confirming its LCST nature (Fig. 6b, c). We next investigated whether LLPSense could guide the reprogramming of this LCST profile toward a UCST-like regime. Adapting a strategy analogous to the α -synuclein mutational screen, we prioritized mutations based on the slope of the difference in predicted LLPS probability between the variant and wild-type (Δ Probability) across the temperature gradient. We selected mutations with the most negative Δ Probability slopes, indicating a shift toward UCST behavior, and incorporated them into the sequence (Fig. 6d). To ensure a sufficient cumulative impact to invert the temperature dependence while safeguarding against loss of phase-separation propensity, mutations were introduced across three iterative rounds, with 10 substitutions accumulated per round (Supplementary Table 4). Characterization of the intermediate variant, UBQLN4-m10 (containing the first 10 substitutions), revealed a turbidity level that remained consistent and largely insensitive to temperature fluctuations, representing a transitional state where the inherent LCST tendency was neutralized (Fig. 6c). This stepwise engineering eventually yielded a

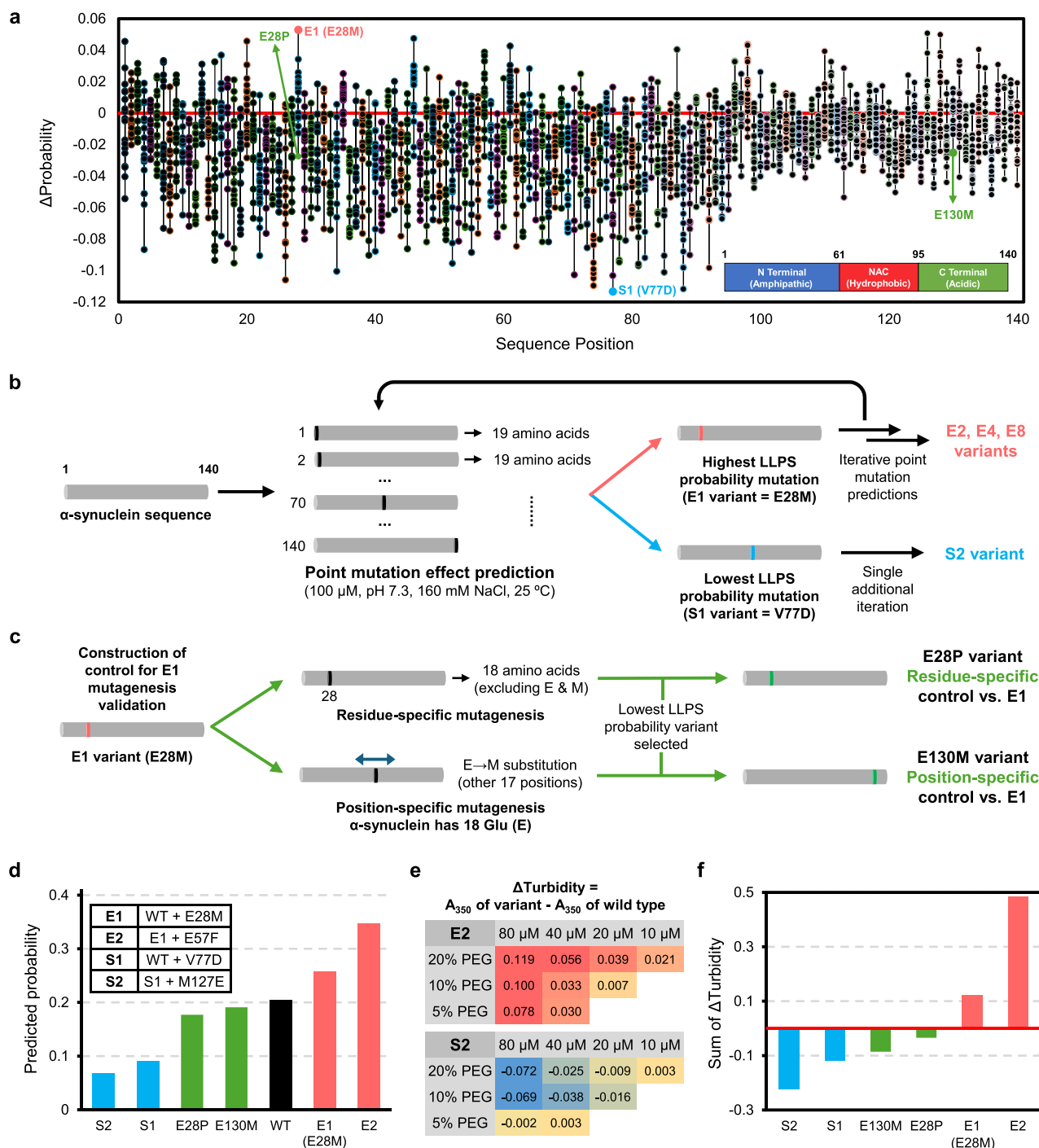


Fig. 4 | Residue-level LLPS propensity mapping of α -synuclein variants using LLPSense. **a** Predicted mutational landscape showing the shift in probability (Δ Probability) for all point mutants relative to the WT (red line), with key variants and controls highlighted. The domain architecture of α -synuclein is shown at the bottom. **b** Schematic workflow of the in silico mutational scanning strategy. E1 (E28M) and S1 (V77D) were identified as the strongest positive and negative modulators of LLPS probability, respectively, serving as scaffolds for recursive multi-mutant design (E2–E8 and S2). **c** Schematic of control variant construction for E1 mutagenesis validation. Controls were designed to decouple position and residue

effects: E28P (same position, different residue) and E130M (different position, same residue). **d** Predicted LLPS probabilities for the selected α -synuclein variants. The inset table details the mutations in each variant. **e** Experimental Δ Turbidity heatmaps for E2 and S2 across protein and PEG8k concentrations, showing phase behavior relative to WT. Warm colors indicate enhancement; cool colors indicate suppression of phase separation. **f** Summed Δ Turbidity across all conditions for the selected variants, with the red line indicating the WT baseline. The experimental trend mirrors the predicted probabilities in (d). Source data are provided as a Source Data file.

UBQLN4-UCST variant that formed condensates resembling percolated clusters (Supplementary Fig. 14b). The recombinantly expressed variant displayed the targeted inversion in temperature dependence, with markedly higher turbidity at lower temperatures than at elevated

temperatures (Fig. 6c). These results demonstrate that LLPSense serves not only as a predictor of condition-dependent phase behavior but also as a tool for tuning it through targeted sequence modifications.

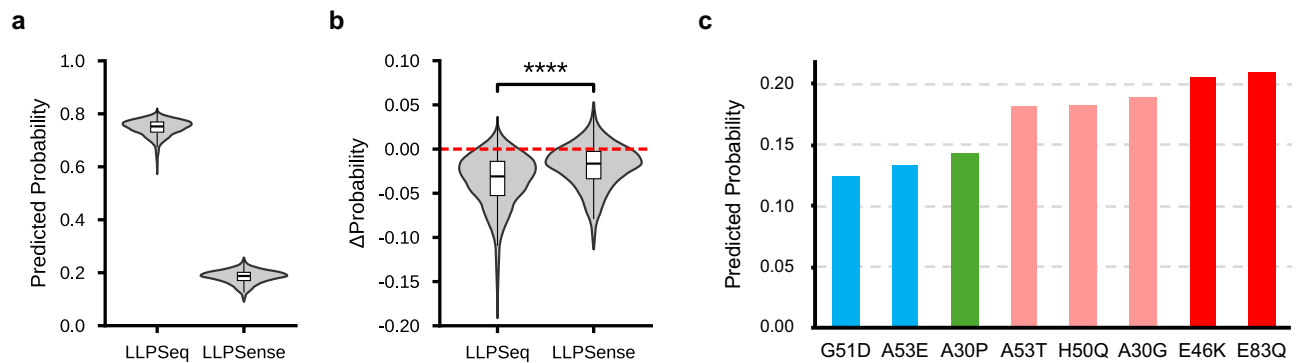


Fig. 5 | Condition-aware screening resolves the bidirectional and fine-grained mutational impacts on α -synuclein LLPS. **a** Violin plots showing the distribution of predicted LLPS probabilities by LLPSeq and LLPSense for the α -synuclein point mutant library ($n = 2,660$; 19 substitutions $\times$ 140 residue positions). **b** Violin plots comparing the Δ Probability distributions of LLPSeq and LLPSense across the same point mutant library ($n = 2660$). Statistical significance in **(b)** was assessed by a two-sided Wilcoxon rank-sum test ($****P = 3.76 \times 10^{-91}$). For both **(a)** and **(b)**, embedded boxplots represent the median (center line), interquartile range (Q1–Q3, box bounds), and whiskers extending to $1.5 \times$ IQR (minima and maxima within this

range); outliers are not shown. **c** Bar plot of predicted LLPS probabilities for eight Parkinson's disease-associated α -synuclein variants, color-coded by literature-reported LLPS propensity: red (significantly enhanced), pink (mildly enhanced), green (comparable), and blue (suppressed) relative to WT. The predicted probabilities show a high qualitative correlation with reported rankings, including (1) $E46K \approx E83Q > A30G \approx A53T > A30P \geq WT > A53E > G51D$, (2) $E46K > H50Q \approx A53T > WT \approx A30P > A53E$, and (3) the identification of E46K, A53T, and H50Q as LLPS-enhancing mutations. Source data are provided as a Source Data file.

Proteome-wide analysis of physicochemical dependencies in phase separation

Having established the robustness of LLPSense, we applied the model to map the global landscape of protein phase separation across the human proteome under diverse physicochemical conditions. Increasing the concentration of macromolecular crowding agents (PEG 8k–20k) induced a proteome-wide shift toward higher predicted LLPS probabilities (Supplementary Fig. 15a). This trend aligns with the excluded volume effect, whereby macromolecular crowding increases the effective protein concentration and promotes condensation. Although these predicted scores may not perfectly correspond to absolute phase separation fractions, they provide a valuable framework for relative comparison, facilitating the benchmarking of the LLPS propensity of proteins of interest against the entire human proteome. On the other hand, increasing ionic strength by elevating NaCl concentrations generally reduced predicted LLPS probabilities (Supplementary Fig. 15c), indicating that electrostatic interactions contribute substantially to condensate formation for a large fraction of human proteins. Interestingly, increasing the NaCl concentration from 500 mM to 1000 mM resulted in a statistically significant increase in predicted probability ($****P < 0.0001$, Wilcoxon tests). This rebound suggests an increased contribution of hydrophobic interactions under ionic strengths far exceeding physiological levels.

We also analyzed the correlation between sequence characteristics and salt sensitivity, quantified by the slope of phase separation probability relative to salt concentration. We observed a negative correlation between this slope and the fraction of charged residues (FCR). This indicates that IDRs with higher overall charge density exhibit a steeper decline in phase separation stability due to electrostatic screening (Supplementary Fig. 16a–b). In contrast, because hydrophobic interactions are typically strengthened at higher ionic strengths, the hydrophathy index exhibited a positive correlation with the salt-response slope (Supplementary Fig. 14c–d). Regarding net charge per residue (NCPR), we found that proteins with a near-neutral net charge ($NCPR \approx 0$) show a higher prevalence of phase separation (Supplementary Fig. 16e). However, these proteins are also relatively sensitive to ionic screening (Supplementary Fig. 16f). Furthermore, it is theoretically expected that LLPS is energetically favored near the pI, as the minimization of electrostatic repulsion facilitates protein–protein interactions. Accordingly, we analyzed the relationship between predicted scores and pH relative to pI (Supplementary Fig. 17). We found that basic proteins (high pI)

sustained their phase separation probabilities even at pH levels exceeding 7.4, whereas acidic proteins (low pI) exhibited a decreasing trend in propensity as conditions became more alkaline. In summary, these results confirm that our model faithfully recapitulates the underlying physicochemical rules governing LLPS across diverse environmental landscapes at the proteome-wide scale.

Discussion

In this study, we presented LLPSense, a unified machine learning framework that bridges intrinsic sequence determinants and extrinsic environmental factors to accurately predict protein phase separation. By harnessing high-dimensional embeddings from pre-trained protein language models, LLPSense overcomes the inherent limitations of previous sequence-only predictors, thereby enabling high-resolution characterization of phase behavior across diverse physicochemical landscapes.

Our comprehensive validation highlights the transformative potential of this condition-aware approach. First, LLPSense identified proteins exhibiting atypical phase-separation behavior, such as SGTA, in datasets previously curated as non-phase-separating. This finding reveals latent inaccuracies in current negative datasets and demonstrates the model's ability to capture complex, reentrant phase behaviors. Secondly, by mapping the mutational landscape of α -synuclein, we demonstrated that the model can bidirectionally predict mutations that modulate phase separation propensity. This capability allows for the rapid identification of high-risk variants with pathogenic potential, effectively prioritizing candidates for experimental follow-up in LLPS-associated pathologies. Lastly, we successfully reprogrammed the thermodynamic profile of UBQLN4 from an LCST to a UCST regime through targeted sequence modification. This serves as a definitive proof-of-concept that LLPSense can facilitate the model-guided design of proteins with tunable, stimuli-responsive phase properties.

As exemplified by the inaccuracies under high salt concentrations in the case of RAD23A, condition-aware LLPS AI still has considerable room for improvement. Detailed precautions for interpreting results with the current version are provided in the Supplementary Notes; we view these present constraints as clear stepping stones for continued advancement. Furthermore, refining sequence-level specificity and addressing potential false-positive inflation under conditions broadly favorable for LLPS remain important topics for subsequent investigation. In this regard, LLPSense's streamlined modular architecture

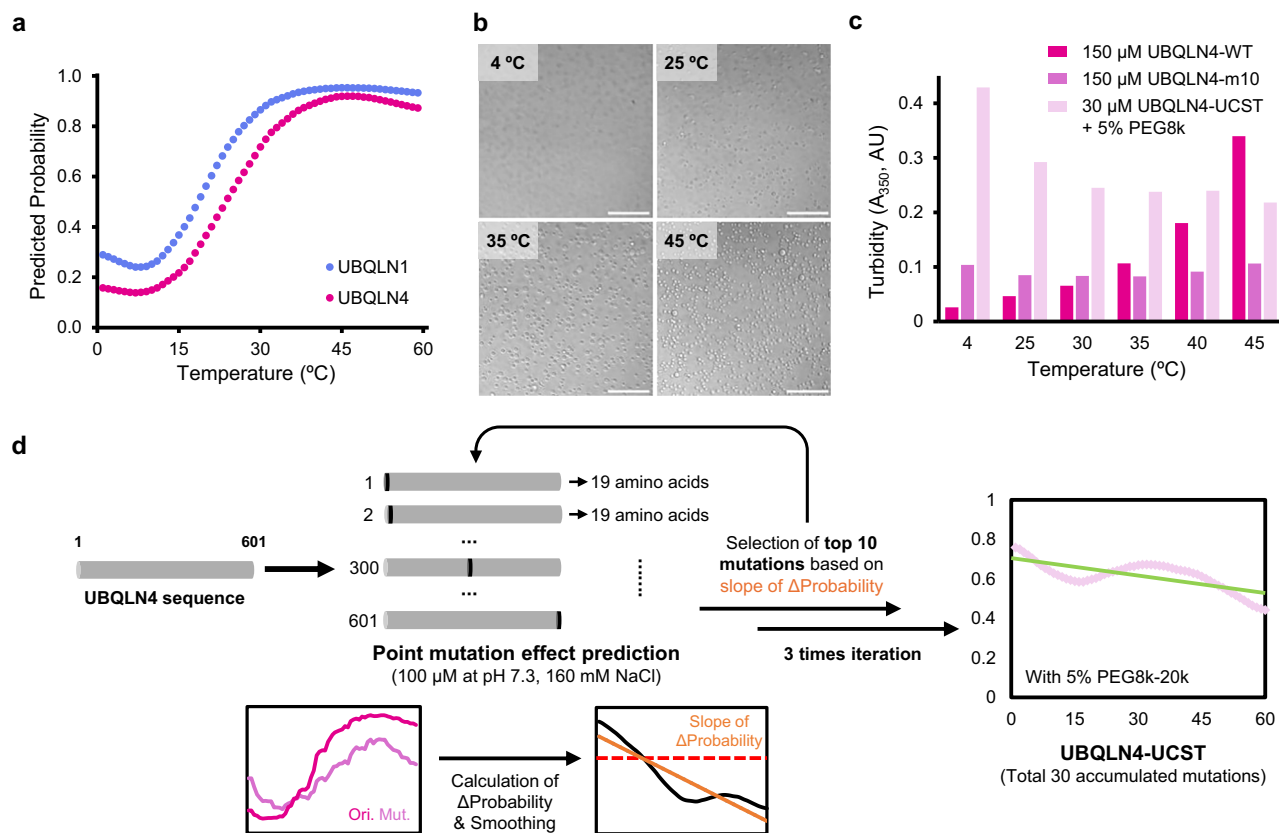


Fig. 6 | Characterization and LLPsense-guided reprogramming of UBQLN4 LCST-to-UCST phase behavior. **a** Predicted temperature-dependent LLPs probability profiles for UBQLN1 and UBQLN4, both exhibiting LCST-type behavior. Predicted points were smoothed using a centered moving average with a 15-point window (15 °C span). **b** Microscopy images of UBQLN4 WT condensates formed across a temperature gradient; each panel shows a representative field from a single experiment. Scale bars, 50 μm. **c** Experimental turbidity comparing the thermal response of UBQLN4 WT, UBQLN4-m10, and UBQLN4-UCST. The progressive shift

from LCST- to UCST-type behavior is observed across the three variants. **d** Schematic of the computational strategy for inverting UBQLN4 phase behavior. The top 10 mutations with the largest ΔProbability slope in the desired direction (sign) were selected per iteration over three rounds, yielding UBQLN4-UCST with 30 accumulated mutations. While the selection is based on individual single-mutation evaluations, the bottom plots show the simultaneous introduction of the top 10 mutations to better visualize the predicted shift in phase behavior. Source data are provided as a Source Data file.

ensures future scalability. As more sophisticated protein language models (pLMs) emerge, LLPsense allows seamless integration of these next-generation representations. Concurrently, the predictive power of LLPsense is expected to grow as more high-quality experimental data are accumulated. This data-driven evolution will extend the framework's applicability beyond binary classification, paving the way for more granular characterizations, such as distinguishing between diverse condensate material states. Building on the foundation of LLPsense, we plan to implement multifaceted strategies, including integrating structural features derived from AlphaFold 3, to develop models with enhanced predictive accuracy and broader generalizability.

By providing a reliable platform for predicting and manipulating condition-dependent phase separation, LLPsense enables rapid, proteome-wide analysis across diverse biological contexts. This versatile capability enables a broad range of investigations, from identifying disease-associated mutations to gaining deeper physical insight into cellular organization and the fundamental principles governing phase behavior. Additionally, this work lays the foundation for programmable biomolecular condensation, opening new avenues for innovation in synthetic biology and therapeutics development through synthetic condensates or by modulating LLPs.

Methods

Methods for the data presented in the Supplementary Information (Support Vector Machine model, naive Bayesian model, SHAP,

dimension reduction of sequence embedding, and per-condition coupling architecture) are detailed therein (Supplementary Methods).

Dataset preparation for LLPSeq training and evaluation

To ensure a fair and consistent comparison, LLPSeq was trained and evaluated using the dataset and evaluation strategy established by the PSPire model. Protein sequences for the PSPire dataset, identified by their UniProt accessions, were retrieved from the UniProt database [<https://www.uniprot.org>] on March 29, 2024. While the positive protein sets (ID-PSPs and noID-PSPs) remained identical to those in the original study, a minor fraction of the negative set (non-PSPs) was excluded. This exclusion was necessary because subsequent UniProt sequence updates altered certain protein lengths, resulting in some exceeding the original PSPire filtering threshold (<2700 amino acids). Specifically, 30 out of 8323 entries in the training set and 2 out of 1,961 entries in the testing set were removed. For the baseline comparison, 55 engineered features were extracted following the methodology described in the DeePhase study⁷. These features were used to train a baseline model to benchmark the performance of our protein language model (pLM)-based LLPSeq.

Sampling of experimental condition data

Entries from LLPSDB v2 were processed into protein names, amino acid sequences, and condition-related variables, including protein concentration, temperature, pH, crowding agents, three types of salts (NaCl, KCl, and MgCl₂), and glycerol percentage. Crowding agents

were classified into six representative categories: PEG 300–1k, PEG 3k–6k, PEG 8k–20k, ficoll, dextran $MW \leq 40$ kDa, and dextran $MW \geq 70$ kDa. PEG and dextran with unspecified molecular weights were assigned to the PEG 3k–6k and Dextran $MW \leq 40$ kDa categories, respectively. Experimental conditions were parsed from a structured Excel dataset into numerical feature vectors. Fields specified as fixed values or ranges ('a-b', 'a-', and '-b') were interpreted according to predefined feature-specific limits, sampling densities, and sampling intervals (Supplementary Table 5). Closed ranges were uniformly sampled either as continuous intervals or as discrete unit-based grids, depending on the feature type. Open-ended ranges were expanded 10-fold above or 0.1-fold below the specified boundary, and subsequently clipped to the global minimum or maximum allowed for each feature. A Cartesian product of all condition values was generated to enumerate every valid experimental configuration. If the resulting combinatorial expansion exceeded a predefined limit ($n > 100$), a random subset of configurations was sampled. Finally, all features were normalized to the range [0, 1] using feature-specific maximum values.

Model training and hyperparameter optimization

The LLPSense model was constructed using the eXtreme Gradient Boosting (XGBoost) framework, a scalable implementation of gradient-boosted decision trees. The model was trained with the 'binary:logistic' objective function to predict the probability of phase separation, minimizing the logarithmic loss (logloss) metric. To ensure robust generalization, we optimized key hyperparameters governing the tree architecture and learning process—specifically `n_estimators`, `max_depth`, `learning_rate`, `min_child_weight`, `subsample`, `colsample_bytree`, `gamma`, and regularization terms (`reg_alpha`, `reg_lambda`). Hyperparameters for both LLPSeq and LLPSense were optimized using the Optuna framework. For LLPSense, model selection was conducted using a 5-fold cluster-based cross-validation scheme. The final optimized hyperparameter configurations for each model are summarized in Supplementary Table 6.

Standard prediction condition and phase-separating candidate screening

Unless otherwise specified, phase-separation probabilities were computed under standard conditions: 100 μ M protein concentration, pH 7.3, 160 mM NaCl, 25 °C, and no other additives. For candidate screening, we employed a temperature-based filtering approach, smoothing predicted probabilities with a centered moving-average (CMA) window of size 9. Candidates were identified as proteins exhibiting a probability exceeding 0.5 within the 10–50 °C range. The complete list of candidates identified through this screening is provided in Supplementary Data 1. For experimental validation, we selected SGTA alongside three randomly drawn proteins (ASCL1, FBLL1, and SEMG2) from a pool of sequences shorter than 1000 amino acids, specifically those exhibiting a predicted phase transition threshold within the 10–50 °C screening window. The AlphaFold database structures⁵² of the proteins used in the experiments are shown in Supplementary Fig. 18.

LLPS prediction of α -synuclein pathogenic mutations and comparison with catGRANULE

We curated LLPS modulation data for α -synuclein mutations from four independent studies. The gathered experimental evidence included mutational rankings derived from turbidity assays—specifically (a) $E46K \approx E83Q > A30G \approx A53T > A30P \geq WT > A53E > G51D$ ⁴⁷ and (b) $E46K > H50Q \approx A53T > WT \approx A30P > A53E$ ⁴⁸—as well as (c) the identification of E46K, A53T, and H50Q as LLPS-enhancing mutations via droplet imaging^{46,49}. By consolidating these prior observations, we categorized the variants into four distinct groups relative to the wild-type (WT): significantly enhanced (E46K, E83Q), mildly enhanced (A30G, A53T, H50Q), comparable to WT (A30P), and suppressed

(A53E, G51D). LLPSense was subsequently employed to predict the LLPS probability for each curated mutant (standard condition). For comparative benchmarking, we used the catGRANULE 2.0 ROBOT web server to generate mutation scores for the same set of variants, as well as for mutations identified at each iterative step of our study.

Model-guided mutagenesis for design of UBQLN4-UCST

We employed an iterative model-guided mutagenesis strategy under standard prediction conditions in the absence of crowding agents. For each round, temperature-dependent probability curves were smoothed using a CMA with a window size of 15. The temperature sensitivity of each single-point mutant was quantified by calculating the slope of the difference in predicted LLPS probability between the variant and wild-type ($\Delta\text{Probability} = \text{Probability}_{\text{variant}} - \text{Probability}_{\text{WT}}$) across the temperature gradient via linear regression. The comprehensive list of these mutations and their calculated $\Delta\text{Probability}$ slopes for each round is provided in Supplementary Table 3. We selected the top 10 mutations with the most negative $\Delta\text{Probability}$ slopes, indicating a shift from LCST to UCST behavior, and incorporated them into the sequence. This procedure was repeated for three consecutive rounds, resulting in a final UBQLN4-UCST variant harboring 30 cumulative mutations. Note that, for visualization in Fig. 5d, the probability curve for UBQLN4-UCST was plotted in the presence of 5% PEG 8k–20k, using a CMA window size of 15 for clarity. The AlphaFold database structure of UBQLN4 WT and the predicted structure⁵³ of UBQLN4-UCST are shown in Supplementary Fig. 18. Finally, although these variants were initially identified using a version of LLPSense prior to its final refinement, the overall predicted temperature-dependent trends and selection logic remain consistent despite minor numerical discrepancies compared to the latest model version (Supplementary Fig. 19).

Expression and purification of proteins

All proteins used in this study were subcloned into the pET-21a(+) vector (Novagen) using the *NdeI* and *XhoI* restriction sites, allowing for expression with the vector-encoded C-terminal His-tag. Complete protein sequences are provided in Supplementary Table 7. The recombinant plasmids were transformed into *Escherichia coli* BL21(DE3) cells and cultured in LB medium supplemented with 100 μ g/mL ampicillin at 37 °C. When the OD_{600} reached 0.4–0.7, protein overexpression was induced by adding 1 mM isopropyl β -D-1-thiogalactopyranoside (IPTG), followed by overnight incubation at 20 °C. Cells were harvested by centrifugation at 6000 rpm ($6400 \times g$) and washed once with 1 $\times$ PBS (160 mM NaCl, pH 7.3). The cell pellets were resuspended in Ni-IDA binding buffer (20 mM Tris-HCl, 150 mM NaCl, 10 mM imidazole, pH 8.0) and lysed by sonication. Lysates were clarified by centrifugation at 12,000 rpm ($16,000 \times g$) for 15 min, and the supernatants were collected. The supernatant was loaded onto a Ni-IDA Excellose (Takara) resin column, washed with wash buffer (20 mM Tris-HCl, 150 mM NaCl, 50 mM imidazole, pH 8.0), and eluted with elution buffer (20 mM Tris-HCl, 150 mM NaCl, 300 mM imidazole, pH 8.0). For α -synuclein variants, the eluate was heat-shocked at 65 °C, then centrifuged at 13,000 rpm ($16,000 \times g$) and filtered through a 0.2 μ m syringe filter (ADVANTEC). Protein solutions were dialyzed against 1 $\times$ PBS at 25 °C, then filtered through a 0.2 μ m syringe filter. For RAD23A, to examine the NaCl concentration-dependent phase separation response, the protein solution was desalted into NaCl-free Tris-HCl buffer (pH 8.0) using a PD-10 desalting column (Cytiva) rather than dialysis. Protein concentrations were determined by measuring absorbance at 280 nm. Final protein solutions were aliquoted, flash-frozen in liquid nitrogen, and stored at –80 °C for long-term use.

Two-dimensional conditional LLPS mapping of LAF-SUMO

Unlike other proteins, the full sequence, including the His-tag, was used for LLPS prediction. Expression conditions were identical to those used for other constructs; however, purification buffers were

modified to accommodate the low-salt instability characteristic of LAF-RGG, using a binding buffer (50 mM Tris-HCl, 500 mM NaCl, pH 8.0), wash buffer (binding buffer plus 50 mM imidazole), and elution buffer (binding buffer plus 500 mM imidazole). For the NaCl-temperature response mapping, both prediction and experimental conditions included 10% glycerol. The purified protein was dialyzed into 1× PBS (160 mM NaCl) supplemented with 10% glycerol. For the pH-NaCl response mapping, the eluted protein was immediately desalted using a PD-10 column (Cytiva) into phosphate buffer (pH 6.0 and 7.0) or Tris-HCl buffer (pH 8.0 and 9.0), each supplemented with 500 mM NaCl. In both mapping experiments, the prepared LAF-SUMO protein solution was adjusted to a final concentration of 50 μ M in the respective buffer conditions for turbidity measurements.

Turbidity assays

Protein phase separation was assessed in 96-well black plates with clear bottoms (Thermo Fisher). Turbidity was quantified by measuring absorbance at 350 nm using a spectral scanning multimode reader (VarioSkan Flash, Thermo Fisher). For each condition, 100 μ L of protein solution was loaded per well. Measurements were recorded 30 min after the induction of phase separation, triggered by either the addition of a crowding agent or a temperature shift. Baseline correction was performed by subtracting the absorbance of PBS control wells containing the corresponding concentration of PEG8k. Each reported turbidity value represents the average of ten consecutive measurements taken from a single sample. In the specific case of α -synuclein variants, fifteen consecutive measurements were recorded per sample. Measurements were designed to identify consistent trends across diverse conditions for the same protein rather than repeated measurements under a single condition. All observed LLPS trends were independently replicated.

Observation of protein condensates via microscopy

To visualize the protein condensates, differential interference contrast (DIC) imaging was performed using a laser-scanning confocal microscope (A1R HD25; Nikon) mounted on an Eclipse Ti2 inverted microscope, equipped with a 60×/1.40 numerical aperture (NA) apochromatic oil-immersion objective. To minimize non-specific surface adsorption, all imaging experiments were performed using BSA-passivated 96-well plates. Surface passivation was conducted by incubating the wells with 0.25% (w/v) BSA in PBS for 2 h at room temperature, followed by twice-washing with PBS to remove residual BSA. Prior to image acquisition, 150 μ L of protein samples were loaded into passivated wells and allowed to settle for 60 min (unless otherwise specified) to ensure droplet formation and equilibration.

Proteome retrieval and physicochemical profiling

We obtained the Swiss-Prot reviewed human proteome from the UniProt database (accessed on November 25, 2025). For the analysis, we specifically selected proteins containing fewer than 4000 residues. These human proteome data were used for the following analyses: probability density plots (Supplementary Figs. 6, 15), violin plots of probability slope distributions (Supplementary Fig. 7), physicochemical profiling (Supplementary Figs. 16, 17), UMAP projections (Supplementary Fig. 20), and SHAP analysis (Supplementary Fig. 23). Sequence-based physicochemical properties, including the fraction of charged residues (FCR), net charge per residue (NCPR), and hydrophathy (0–9 scale), were calculated using the sparrow library (<https://sparrow-online.com>).

Implementation details

All computational experiments were conducted using a single NVIDIA RTX A6000 GPU. Feature extraction utilizing the ProtT5 model required ~131 ms per protein on average (benchmarked on the human proteome restricted to lengths under 4000 residues). For inference,

LLPSense processed a single protein-condition pair in approximately 34 ms. Notably, inference consumed less than 8 GB of VRAM, demonstrating the framework's computational efficiency for large-scale screening.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Unless otherwise stated, all data supporting the results of this study can be found in the article, supplementary, and source data files. The pre-established models, curated dataset, and input template file are publicly available on GitHub (<https://github.com/NearNiah/LLPSense>). The dataset was curated based on LLP SDB v2.0 (<http://bio-comp.org.cn/llpsdb>). Protein sequences for the PSPire dataset were retrieved from the UniProt database (<https://www.uniprot.org>) on March 29, 2024, using the UniProt accessions specified in the original PSPire publication. The Swiss-Prot reviewed human proteome used for proteome-wide analyses was obtained from the UniProt database under proteome accession UP000005640 (accessed on November 25, 2025). Uncropped gel scan and photographic image for Supplementary Fig. 11 are available in a separate repository (<https://doi.org/10.5281/zenodo.21208572>). The complete raw data underlying SHAP analysis are provided as Supplementary Data 2–3. Source data are provided with this paper.

Code availability

The LLPSense and LLPSeq codes developed in this study are available on GitHub (<https://github.com/NearNiah/LLPSense>). The version used in this study has been archived on Zenodo (<https://doi.org/10.5281/zenodo.21206758>)⁵⁴.

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Author contributions

J.B. and M.K. conceived the study and developed the machine learning model and codebase (Conceptualization, Methodology, Software). D.L. curated the datasets under the guidance of J.B. (Data curation). J.B. performed the wet-lab experiments with support from D.L. (Investigation). M.K. performed the computational analysis and evaluated model

performance (Formal analysis, Validation). J.B. conducted the experimental data analysis, visualized the results (Formal analysis, Visualization), and drafted the manuscript (Writing—original draft). Y.J. and M.K. reviewed and edited the manuscript (Writing—review & editing). Y.J. provided guidance on experimental design and data interpretation and, together with K.-J.Y., supervised the project (Supervision). K.-J.Y. and Y.J. acquired funding for the study (Funding acquisition).

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Competing interests

Y.J., J.B., M.K., and K.-J.Y. are inventors on a patent application filed by the Korea Advanced Institute of Science and Technology (KAIST; Application No. KR 10-2026-0068712) entitled “Method and apparatus for predicting condition-dependent protein phase separation and designing sequences,” which covers the machine learning framework (LLPSense) described in this manuscript. D.L. declares no competing interests.

Additional information

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